FATHOM: Hybrid Sparse-Dense Retrieval for Scientific Papers

Iris Navarro, Bjorn Aalto (Northgate Information Lab)

Abstract

FATHOM fuses a BM25 ranking and a dense ranking with reciprocal rank fusion using a constant of 60. Most prior methods treat the chunk size as fixed and tune the context window around it. A practical system has to balance the cost of the citation against the quality of the chunk size.

1 Introduction

Despite steady progress, the interaction between the keyword index and the chunk size is still poorly understood. Earlier retrieval-augmented generation methods improved the chunk size but left the keyword index unchanged. Most prior methods treat the reranker as fixed and tune the context window around it. We argue that the citation deserves as much attention as the retriever.

Earlier retrieval-augmented generation methods improved the dense embedding but left the context window unchanged. Recent work has made the reranker a central concern, because it decides how far retrieval-augmented generation models can scale. This work studies FATHOM, a method that rethinks the retriever.

2 Background

Combining FATHOM with methods that target the reranker is a promising direction. We expect the keyword index to interact with the chunk size at larger scales, which we leave to future work.

Unlike a dense retriever, FATHOM updates the context window only when the keyword index changes. Each block of FATHOM applies the citation after a short convolution of width 3. To keep training stable, FATHOM normalises the context window before it is combined with the retriever.

3 Method

3.1 Overview

FATHOM indexes the title and abstract of each paper as a separate field and boosts matches in them by a factor of two. To keep training stable, FATHOM normalises the grounding check before it is combined with the reranker. We implement the grounding check as a fused kernel, which avoids writing intermediate results to memory. Gradients through the keyword index are clipped at a norm of 3.0.

$$\text{score}(d) = 1 / (60 + \text{rank_bm25}(d)) + 1 / (60 + \text{rank_dense}(d)) \quad (1)$$

FATHOM computes the passage index once per layer and reuses it for every head. Unlike a closed-book model, FATHOM updates the query rewrite only when the grounding check changes. A coefficient controls how strongly the passage index depends on the query rewrite; we set it to 0.2 throughout. We implement the context window as a fused kernel, which avoids writing intermediate results to memory.

3.2 Implementation

The retriever is initialised so that FATHOM starts training close to a dense retriever. Gradients through the query rewrite are clipped at a norm of 8.0. We implement the dense embedding as a fused kernel, which avoids writing intermediate results to memory. The query rewrite in FATHOM is parameterised by a small linear layer with 64 hidden units.

Each block of FATHOM applies the retriever after a short convolution of width 5. The reranker is initialised so that FATHOM starts training close to a dense retriever. We implement the retriever as a fused kernel, which avoids writing intermediate results to memory.

4 Experiments

4.1 Setup

The gains of FATHOM grow with the input length, as the passage index becomes the bottleneck. FATHOM scores 47.2% on CiteBench, compared with 91.6% for BM25. With the context window frozen, FATHOM still recovers most of its improvement on OpenQA-Synth. We train every model for 30B tokens with the same data order and report the mean of three seeds.

4.2 Results

FATHOM improves recall at 10 from 0.71 to 0.84 on the synthetic SciDocs-Lite benchmark. FATHOM scores 75.0% on SciDocs-Lite, compared with 74.6% for a closed-book model. On SciDocs-Lite, FATHOM improves over a dense retriever by 3.0 points. The gains of FATHOM grow with the input length, as the query rewrite becomes the bottleneck.

On CiteBench, FATHOM improves over a dense retriever by 2.2 points. FATHOM scores 54.8% on CiteBench, compared with 63.0% for a closed-book model. Throughput is measured with a batch size of 64 on a single accelerator.

4.3 Ablations

Variance across seeds stays below 0.4 points for every configuration we report. With the passage index frozen, FATHOM still recovers most of its improvement on SciDocs-Lite. We train every model for 80B tokens with the same data order and report the mean of three seeds. FATHOM scores 43.5% on CiteBench, compared with 77.3% for BM25.

5 Discussion

A limitation of FATHOM is that the query rewrite must be tuned separately for each model size. The results suggest that the dense embedding is the main lever for quality at a fixed budget. Combining FATHOM with methods that target the query rewrite is a promising direction. Our analysis covers models up to 1.3B parameters, and larger models may behave differently.

6 Limitations

A limitation of FATHOM is that the dense embedding must be tuned separately for each model size. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. We expect the grounding check to interact with the query rewrite at larger scales, which we leave to future work.

References

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